RSIC-GMamba: A State-Space Model With Genetic Operations for Remote Sensing Image Captioning

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***Abstract*— Recent advancements indicate that the novel Mamba framework, with its linear reasoning capabilities and performance on par with the Transformer framework, has become a popular alternative to Transformers. However, there is still a lack of exploration in remote sensing image captioning (RSIC). One important challenge is that the 1-D selective state-space model (SSM) is not suitable for 2-D remote sensing images (RSIs). To mitigate this challenge, this article proposes an SSM with genetic operations for RSIC (RSIC-GMamba) that integrates the search capabilities of heuristic genetic algorithm and the global modeling capabilities of Transformer into the Mamba framework. To comprehensively capture the multiscale global context of RSIs, we design a genetic SSM by incorporating a dilated convolution, genetic operations (crossover and mutation), and self-attention mechanism into the selective SSM. Specifically, dilated convolutions are employed to extract multiscale visual features through varying dilation rates. Crossover operation expands the spatial arrangement of image regions to thoroughly capture contextual information and mutation operation introduces randomness to enhance the model's robustness. The self-attention mechanism is integrated to model relationships among SSM hidden states, thereby enhancing visual context. In addition, to fully utilize high- and low-level visual-semantic information, we propose a vision and scene text aggregation (ViSTA) module based on gating mechanisms. Experimental results on four RSIC datasets demonstrate the effectiveness of the proposed RSIC-GMamba. The code will be publicly available at https://github.com/One-paper-luck/RSIC-GMamba.**

***Index Terms*— Genetic, Mamba, remote sensing image captioning (RSIC), scan mechanism.**

Received 11 October 2024; revised 8 January 2025 and 15 February 2025; accepted 3 March 2025. Date of publication 6 March 2025; date of current version 25 March 2025. This work was supported in part by Jiangsu Geological Bureau Research Project under Grant 2023KY11, in part by Jiangsu Key Laboratory of Spectral Imaging and Intelligent Sense under Grant JSGP202101 and Grant JSGP202204, in part by the Frontier Technologies Research and Development Program of Jiangsu under Grant BF2024070, in part by the National Natural Science Foundation of China under Grant 62471235 and Grant 62302255, and in part by China Postdoctoral Science Foundation under Grant 2023TQ0181. *(Corresponding authors: Liang Xiao; Jing Wang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3548664

I. Introduction

REMOTE sensing image captioning (RSIC) aims to describe the fine-grained visual content in remote sensing images (RSIs) using natural language. Historically, numerous methods [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] based on the encoder–decoder paradigm have been proposed for natural image captioning (NIC). Building on the NIC methods, research in RSIC has evolved from the convolutional neural network (CNN)-recurrent neural network (RNN) framework [16], [17], [19], [20], [21], [22], [23], [24], [25], [26], [27] to the more advanced Transformer framework [28], [29], [30], [31], [32]. In contrast to RNN, Transformer calculates the relevance between each token in the input sequence and every other token using a self-attention mechanism. This architecture not only captures long-range dependencies with ease but also supports efficient parallel computing. However, it is important to note that Transformer demands significantly more computational resources and storage space. Their runtime increases quadratically with the sequence length, which starkly contrasts with the linear time complexity of RNNs. In addition, Transformer models are highly sensitive to the global contextual information of the input data, where even minor changes can have a substantial impact on the final output.

Recent studies have shown that RNN-like state-space models (SSMs) [33], [34], [35], [36], [37] possess effective long-sequence modeling capabilities, linear computational complexity, and efficient parallel computation, positioning them as alternatives to Transformers. A series of Mamba variants [38], [39], [40], [41], [42] integrate selective SSMs and multilayer perceptrons (MLPs), achieving performance comparable to that of Transformers in both visual tasks and long text generation tasks. However, the application and exploration of Mamba in multimodal tasks are relatively limited. This may be due to the fact that the selective scanning mechanism is primarily designed for 1-D causal sequences, resulting in significant shortcomings when modeling the contextual relationships of 2-D images. To mitigate this issue, recent works [39], [40], [41], [42] have been devoted to investigating 2-D scanning mechanisms for natural images. They predefined multiple scanning directions through several SSM branches to fully capture the dependencies between image regions. However, these methods still face challenges when dealing with RSIs. Natural images are typically captured by cameras from a horizontal perspective, in accordance with human habits of observing scenes, and their visual features are predominantly distributed along the horizontal and

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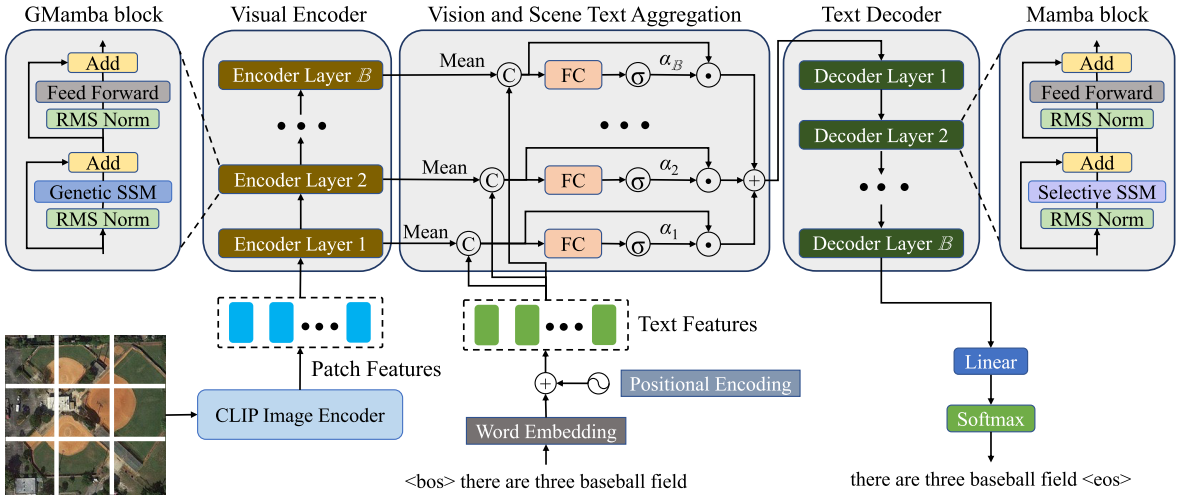


Fig. 1. Framework of the proposed RSIC-GMamba in this article. A pretrained CLIP image encoder is employed to extract patch features from an RSI. Then, the features are fed into a visual encoder. Each encoder layer utilizes a GMamba block. The GMamba block adopts genetic SSMs and FFNs to capture contextual information of visual features. The ViSTA module fuses text features and visual features from high layer and low layer. The fused features are fed into a text decoder to generate words one by one. Each decoder layer employs a vanilla Mamba block. ©, ⊙, σ, and ⊕ represent the concatenation operation, Hadamard product, sigmoid activation function, and matrix multiplication, respectively.

vertical axes. When natural images are rotated, it can affect semantic information. In contrast, RSIs obtained from a bird's-eye view exhibit characteristics of rotational angle ambiguity, for instance, an RSI featuring airplanes at various rotation angles. Therefore, specific scanning directions are insufficient to adequately capture the contextual information of RSIs. A scanning direction can be viewed as a permutation of image regions. The most straightforward approach is to set up N! SSM branches to traverse the N! scanning directions corresponding to N image regions. This undoubtedly leads to a significant increase in the number of parameters. We propose incorporating crossover and mutation operations from the heuristic genetic algorithm [43] to reduce redundant SSM branches. The crossover operation effectively expands the permutation and combination space of image regions, while the mutation operation introduces randomness to enhance the model's robustness. In addition, many works [44], [45] endeavor to integrate the strengths of SSM and Transformer to enhance context modeling capabilities. Building on the vanilla Mamba framework, Lieber et al. [44] stacked Mamba layers and Transformer layers to fully leverage the benefits of both frameworks. Pilault et al. [45] designed a block-state Transformer layer, which adds a context-aware SSM sublayer before each Transformer sublayer. However, the significant architectural differences between Mamba and Transformer increase the complexity of training and hyperparameter tuning when hybridized. Therefore, we recommend embedding a lightweight self-attention layer within the SSM layer to further enhance the modeling capability of the SSM hidden states.

Moreover, the widely used meshed cross-attention in Transformers [12], [30], [31], [32] can effectively aggregate semantic information from both high- and low-level features, facilitating intermodal interactions. However, it cannot be directly integrated into the Mamba framework, as its masking mechanism may disrupt the linear reasoning capabilities of the Mamba decoder. Therefore, we designed a vision and scene text aggregation (ViSTA) module based on a gating mechanism to better suit the Mamba framework.

Based on the aforementioned idea of expanding the arrangement of image regions and incorporating self-attention modeling capabilities into SSM, we propose an RSIC-GMamba for RSIC, as shown in Fig. 1. The proposed framework consists of three main components: a visual encoder, a ViSTA module, and a text decoder. Each encoder layer utilizes a GMamba block, while each decoder layer employs a standard Mamba block. To incorporate prealigned image–text information from large visual–language models, we employ pretrained contrastive language-image pretraining (CLIP) [46] image encoder to extract RSI features for an RSI. The encoder employs genetic SSM to capture contextual information of multiscale visual features. The ViSTA module fuses visual features from all encoding layers and text features. The fused features are then input into the text decoder to generate sentences for the RSI.

The main contributions of this work can be summarized as follows.

1) To expand the arrangement space of image regions, we propose a genetic SSM that integrates genetic operations (i.e., crossover and mutation operations), and self-attention mechanism into the selective SSM, thereby enhancing the capability of the SSM in processing 2-D RSIs.
2) To enhance the model's ability to understand image content, we design a ViSTA module based on gating mechanisms and mean operations to aggregate multilevel visual features and textual features.
3) Extensive experiments are conducted on three RSIC datasets, demonstrating the superior performance of the proposed method.

II. Related Work

In this section, we will review the related works from three aspects: SSM, NIC, and RSIC.

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A. State-Space Model

SSM [33] originated in the field of system control, revealing the connections among internal system states, external inputs, and outputs. Due to its ability to scale linearly with the length of the sequence, it has garnered significant attention in long sequence modeling tasks. To address long-range dependency issues, Gu et al. [34], [35] proposed a structured state-space sequence model (S4), which imposes low-rank constraints on the state matrices to improve computational efficiency. Furthermore, parallel scanning [36], [37] facilitates the parallel training of SSMs. However, the aforementioned approaches cannot perform content-based reasoning. Gu and Dao [38] proposed a selective SSM to address the time-invariant limitations of SSMs. They integrated the selective SSM with MLP block of Transformer into a unified Mamba framework Mamba, achieving state-of-the-art (SOTA) performance in language, audio, and genomics modalities.

Nowadays, the research on Mamba primarily focuses on visual tasks. To bridge the gap between 1-D causal scanning mechanism and 2-D image processing, Zhu et al. [39], [47] proposed a Vision Mamba (Vim) that designs two SSM branches to process forward and backward patch sequences, respectively. Later, Liu et al. [40] proposed a Vmamba model that adopts the 2-D-selective-scan module to aggregate four SSM scanning branches. Yang et al. [41] adopted zigzag scanning to ensure spatial continuity. To enhance the local modeling capability of the model, Huang et al. [42] proposed a local scanning strategy that performs multidirectional scanning with SSMs on tokens within each image window. Lieber et al. [44] utilized the mix of experts architecture to fully leverage the advantages of both Mamba and Transformer frameworks.

B. Natural Image Captioning

With the advancement of deep learning, Vinyals et al. [1] first introduced a vanilla encoder–decoder framework for NIC, utilizing a CNN to encode visual content and an RNN to decode words in an autoregressive manner. Subsequently, Xu et al. [2] incorporated an attention mechanism that dynamically focuses on the visual regions relevant to each word. Following this, the integration of semantic attributes [3], [4] that complement visual content further enhanced the model's visual understanding. In contrast to the visual representations used in the aforementioned methods, employing object detectors [5], [6], [7] to extract regions of interest enables more accurate descriptions. Huang et al. [8] introduced a self-attention mechanism based on the traditional attention mechanism to focus on the correlation between attention outputs and queries. In addition, to address the exposure bias problem resulting from cross-entropy training, Rennie et al. [9] proposed a self-critical training strategy (SCST) based on the reinforcement learning (RL) to facilitate NIC. To address the reward hacking issue, Li et al. [58] proposed a meta-learning method that leverages supervision from ground truth (GT) while optimizing the reward function within RL. To assist models in comprehending new visual concepts that emerge in test sets, Cao et al. [59] introduced a feature deformation meta-network for describing novel objects.

Benefiting from the success of Transformer in natural language processing, various Transformer-based methods for NIC have emerged. Guo et al. [10] combined the relative geometric relationships between objects with attention maps. Subsequently, Zhang et al. [11] validated the effectiveness of capturing positional relationships between pixels. Cornia et al. [12] encoded prior relationships between objects and employed a meshed cross-attention to aggregate multilayer visual features. Pan et al. [13] enhanced model efficiency and performance through linearized attention computations. The advent of pretrained vision–language large models [14], [15] has significantly advanced NIC.

C. Remote Sensing Image Captioning

Similar to the development of NIC, RSIC has transitioned from CNN-RNN-based methods to Transformer-based methods. Qu et al. [19] initially explored the optimal combination of CNN and RNN for RSIC. Lu et al. [20] introduced an attention mechanism to achieve better performance. To generate descriptions suited to remote sensing scenes, Zhang et al. [21] proposed an attribute attention mechanism to address the "semantic gap" between low-level features and high-level semantics in RSIs. Zhang et al. [22] used the word embedding vectors of predicted categories to guide the computation of attention masks. Ye et al. [23] designed an attribute-guided decoder to facilitate word generation. Zhang et al. [24] filtered features weakly correlated with global visual features through an attention gate. Li et al. [25] captured visual-semantic context with enhanced cross-modal semantic perception through two parallel attention branches. To tackle multiscale challenges, Wang et al. [26] fused the features of the last two convolutional layers of CNNs using cross-attention. Li et al. [27] used adaptive average pooling operations to process multiscale features extracted by the efficient spatial pyramid (ESP) [48]. Ni et al. [60] integrated an object counting module into the captioning model to generate accurate quantifiers. Yang et al. [61] utilized hierarchical feature aggregation and cross-modality feature alignment to generate high-quality captions. Cheng et al. [62] utilized a hierarchical attention mechanism to dynamically integrate visual features across various spatial areas and levels of detail, while incorporating a contextual attention component to uncover the underlying, implicit contexts present in RSIs. Recently, Li et al. [74] proposed a novel underwater image captioning task and conducted preliminary explorations by integrating an object detector with an encoder–decoder framework.

Due to the exceptional performance of Transformer on image captioning, Transformer-based methods for RSIC are gradually emerging. Wang et al. [28] extracted meaningful words through a classification task and employed a Transformer to arrange these words into coherent sentences. Following the general procedure of NIC, Zhuang et al. [29] initially trained a Transformer model using cross-entropy loss to establish a robust baseline. Subsequently, they enhanced the model's performance by employing the SCST. Meng et al. [30] incorporated prior scene–object knowledge into self-attention mechanism guiding the model to focus on scene-relevant objects. Liu et al. [31] integrated multiscale features extracted by CNNs and inputted them into the

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visual encoder. Subsequently, they utilized long short-term memory networks (LSTMs) to aggregate information from all encoding layers. Wang et al. [71] leverages a CNN to extract both global and local visual features, while the Transformer decoder bolsters the model's interpretability via self-attention and co-attention mechanisms. Yang et al. [63] proposed a multigranularity instruction multimodal model for attribute-guided RSI detailed description and constructed a corresponding dataset to validate its effectiveness. The methods of zero-shot transfer and fine-tuning multimodal large models further promote RSIC. Meng et al. [32] designed a semantic correlation module to integrate CLIP image embeddings and region features and proposed a global grouping attention mechanism to enhance the local modeling capability of self-attention. Yang et al. [64] proposed a two-stage vision–language pretraining model named BITA, which aligns RSI and text features using a lightweight interactive Fourier Transformer to enhance semantic consistency in RSIC.

Despite the significant advancements of the Transformer in RSIC, its quadratic time complexity leads to relatively low efficiency during inference. In this article, we propose to integrate the strengths of the Transformer into Mamba to achieve more efficient computation.

III. Method

The objective of the proposed RSIC-GMamba is to integrate the novel selective SSM (i.e., Mamba) into RSIC. This section commences with an introduction to the foundational concept of SSM. Subsequently, it presents a comprehensive overview framework of RSIC-GMamba, elucidating its structural components and operational flow.

A. Preliminaries

1) State-Space Model: SSM is a linear sequence model, similar to an RNN. For a continuous system, it transforms a 1-D function or sequence $x(t) \in \mathbf{R} \rightarrow y(t) \in \mathbf{R}$ into an implicit latent state $h(t) \in \mathbf{R}^N$

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) \tag{2}$$

where $\mathbf{A} \in \mathbf{R}^{N\times N}$, $\mathbf{B} \in \mathbf{R}^{N\times 1}$, and $\mathbf{C} \in \mathbf{R}^{1\times N}$ are projection matrices.

2) Structured State-Space Sequence Module (S4): In order to adapt to deep learning computational frameworks, S4 utilizes the zero-order hold (ZOH) method to discretize the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ with a timescale parameter Δ, resulting in the discrete parameters $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$, defined as follows:

$$\overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \tag{3}$$

$$\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}. \tag{4}$$

The discretized SSM can be computed as follows:

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \tag{5}$$

$$y_t = \mathbf{C}h_t. \tag{6}$$

Mathematically, the model facilitates efficient parallel computation through global convolution as follows:

$$y = x * \overline{\mathbf{K}} \tag{7}$$

$$\overline{\mathbf{K}} = \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right) \tag{8}$$

where $*$ represents the convolution operation and $\overline{\mathbf{K}} \in \mathbf{R}^L$ is a structured convolution kernel.

3) Selective SSM (S6): To address the limitations in context-based reasoning imposed by the linear time-invariant property of SSMs, Mamba introduces a selection mechanism that makes parameters $\mathbf{B}$, $\mathbf{C}$, and Δ dependent on the input x, as detailed in the following:

$$\mathbf{B}, \mathbf{C}, \Delta = \text{Linear}(x) \tag{9}$$

where $\mathbf{B} \in R^{B\times L\times N}$, $\mathbf{C} \in R^{B\times L\times N}$, and $\Delta \in R^{B\times L\times D}$. B and D represent the batch size and the number of channels, respectively. To achieve efficient parallel training, S6 employs a parallel scanning (pscan) mechanism [38] to calculate (6) and obtain the hidden state.

4) Mamba: Similar to Transformer architecture, Mamba stacks repetitive Mamba blocks, each succeeded by a normalization layer and a residual connection. Within each Mamba block, an H3 [49] and a gated MLP are seamlessly integrated.

B. Overall Framework

The overall framework of the proposed RSIC-GMamba is shown in Fig. 1. It consists of a visual encoder, a ViSTA module, and a text decoder. Specifically, a set of patch features are first extracted from an RSI using a pretrained CLIP image encoder. Then, the encoder employs genetic SSMs to capture the relationships between different visual features. The ViSTA module aggregates visual features of all encoder layers and text features. Finally, the aggregated feature is fed into the decoder to generate captions for the RSI.

C. Feature Extraction

Following a standard practice in RSIC [32], we adopt a pretrained CLIP image encoder to extract L patch features $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_L\} \in \mathbf{R}^{L\times d_1}$. Then, the features are projected into a d-dimensional space as follows:

$$\mathbf{X}_f = \mathbf{X}\mathbf{W}_f + \mathbf{b}_f \tag{10}$$

where $\mathbf{W}_f \in \mathbf{R}^{d_1\times d}$ and $\mathbf{b}_f \in \mathbf{R}^d$ are the learnable parameter matrix and the bias, respectively. Subsequently, a ReLU activation function and a layer normalization operation are applied. It serves as the input for the visual encoder.

D. Genetic SSM

The 1-D selective SSM of Mamba is not well suited for handling 2-D RSIs. To prevent a dramatic increase in model parameters resulting from the establishment of excessive SSM branches, we propose a genetic SSM, as illustrated in Fig. 2. Compared to the selective SSM, the genetic SSM introduces a dilated convolution and designs mutation and crossover branches. It obtains multiscale visual features through various dilated convolutions. The mutation and crossover branches

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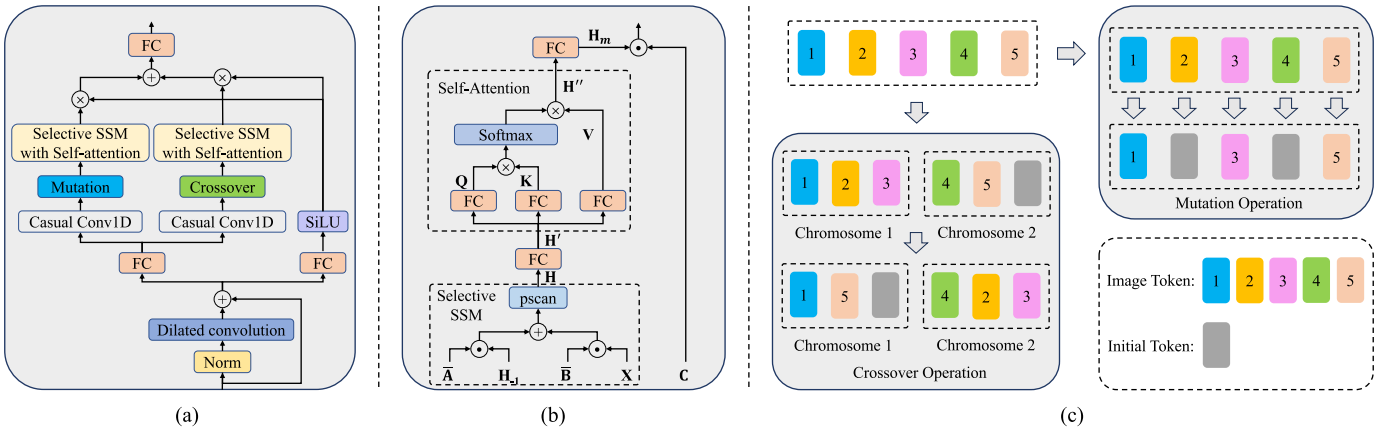


Fig. 2. (Left) Genetic SSM includes the mutation branch and the crossover branch. (Right) Detailed depiction of the mutation operation and crossover operation within the two branches. $\odot$, $\oplus$, and $\otimes$ represent the Hadamard product, addition operator, and matrix multiplication, respectively. (a) Genetic SSM. (b) Selective SSM with self-attention (S7). (c) Crossover and mutation operations.

introduce genetic operations (i.e., mutation and crossover) to fully capture the contextual information of 2-D RSIs and further utilize the self-attention mechanism to model the dependencies of the hidden states in the selective SSM. The details are as follows.

Given the input image feature $\mathbf{X}_I \in \mathbf{R}^{L\times d}$, we first adopt a dilated convolution to expand the receptive field

$$\mathbf{X}'_I = \text{Linear}(\text{DC}(\text{Norm}(\mathbf{X}_I)) + \mathbf{X}_I) \tag{11}$$

where $\mathbf{X}'_I \in \mathbf{R}^{L\times ed}$, e is the expand operator, DC denotes the dilated convolution, and Norm refers to the layer normalization operation. Subsequently, the obtained features are fed into the crossover and mutation branches.

1) Mutation Branch: The core of the mutation branch is a mutation operation and an S6 with self-attention (S7). The mutation operation is depicted on the right of Fig. 2. Drawing an analogy from biological gene mutation, we can consider each image token as a gene. Inspired by genetic algorithm [43], each token is replaced with an initial token drawn from a normal distribution $\mathcal{N}(0, 1)$ if $\eta < P_m$. η is a random number between (0, 1) and $P_m \in [0, 1]$ represents the mutation probability. By introducing randomness, the mutation branch can achieve a regularization effect that enhances the model's learning capacity and generalization ability. Subsequently, the mutated image tokens are input into the S7. S7 incorporates self-attention into S6, which employs a self-attention mechanism and two linear layers on the hidden state matrix $\mathbf{H} \in \mathbf{R}^{L\times ed\times N}$. $\mathbf{H}$ is determined using (3)–(5) and (9), along with calculations derived from the pscan process. Through this operation, the model's contextual modeling capability is further enhanced. The details are as follows:

$$\mathbf{H}' = \text{Linear}(\text{Reshape}(\mathbf{H})) \tag{12}$$

$$\begin{aligned}\mathbf{H}'' &= \text{Att}\left(\mathbf{H}'\mathbf{W}_q, \mathbf{H}'\mathbf{W}_k, \mathbf{H}'\mathbf{W}_v\right)\\ &= \text{softmax}\left(\frac{\left(\mathbf{H}\mathbf{W}_q\right)\left(\mathbf{H}'\mathbf{W}_k\right)^{\mathbf{T}}}{\sqrt{d_m}}\right) \times \left(\mathbf{H}'\mathbf{W}_v\right)\end{aligned} \tag{13}$$

$$\mathbf{H}_m = \text{Linear}\left(\text{Reshape}\left(\mathbf{H}''\right)\right) \tag{14}$$

where $\mathbf{H}' \in \mathbf{R}^{L\times d_m}$, $\mathbf{H}'' \in \mathbf{R}^{L\times d_m}$, and $\mathbf{H}_m \in \mathbf{R}^{L\times ed\times N}$; and $\mathbf{W}_q \in \mathbf{R}^{d\times d_m}$, $\mathbf{W}_k \in \mathbf{R}^{d\times d_m}$, and $\mathbf{W}_v \in \mathbf{R}^{d\times d_m}$ are learnable parameter matrices. By applying (6), we obtain the output of S7. The complete mutation branch can be represented as follows:

$$\mathbf{Y}_m = \text{S7}\left(\text{Mutation}\left(\text{SiLU}\left(\text{Conv1D}\left(\text{Linear}\left(\mathbf{X}'_I\right)\right)\right)\right)\right) \tag{15}$$

$$\mathbf{Z} = \text{SiLU}\left(\text{Linear}\left(\mathbf{X}'_I\right)\right) \tag{16}$$

$$\mathbf{X}_m = \mathbf{Z} \odot \mathbf{Y}_m \tag{17}$$

where Conv1D denotes the 1-D convolution, SiLU is an activation function, and $\odot$ refers to the Hadamard product.

2) Crossover Branch: Similar to the mutation branch, crossover branch consists of a crossover operation and an S7. The crossover operation is inspired by the phenomenon of chromosomal crossover in biology. As shown on the right of Fig. 2, L image tokens are divided into λ groups in order. Each group includes $\lceil L/\lambda \rceil$ tokens, which is considered as a chromosome. If $L \notin \lambda\mathbb{Z}$ ($\mathbb{Z}$ represents the set of all integers), the last chromosome will be padded with $(\lceil L/\lambda \rceil - L\%\lambda)$ initial tokens. Symbol $\lceil\ \rceil$ denotes the ceiling function, which rounds a number up to the nearest integer. Symbol % denotes the modulo operation, representing the remainder of a division. Subsequently, any two chromosomes exchange their last c tokens if $\gamma < P_c$. γ is a random number between (0, 1) and $P_c \in [0, 1]$ represents the crossover probability. Finally, the crossed image tokens are fed into S7. The complete crossover branch is indicated as follows:

$$\mathbf{Y}_c = \text{S7}\left(\text{Crossover}\left(\text{SiLU}\left(\text{Conv1D}\left(\text{Linear}\left(\mathbf{X}'_I\right)\right)\right)\right)\right) \tag{18}$$

$$\mathbf{X}_c = \mathbf{Z} \odot \mathbf{Y}_c. \tag{19}$$

Note that we need to restore the forward order and remove the initial tokens filled in $\mathbf{X}_c$ to satisfy (20). The output of genetic SSM can be calculated as follows:

$$\text{GSSM}(\mathbf{X}_I) = \text{Linear}(\mathbf{X}_m + \mathbf{X}_c). \tag{20}$$

For convenience, we present the operations of genetic SSM in Algorithm 1.

E. RSIC-GMamba

The proposed RSIC-GMamba is composed of a visual encoder, a ViSTA module, and a text decoder. The encoder extracts multiscale visual features with dilated convolutions. After that, it transforms these features into a series of hidden

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Algorithm 1 Genetic SSM

Require: patch features $\mathbf{X}_I$:(B,L,d)
Ensure: patch features $\mathbf{X}_{I+1}$:(B,L,d)
$\mathbf{X}'_I$: (B,L,d) $\leftarrow$ Linear(DC(Norm($\mathbf{X}_I$)) + $\mathbf{X}_I$)
$\mathbf{Z}$: (B,L,ed) $\leftarrow$ SiLU(Linear($\mathbf{X}'_I$))
/* process with different SSM branches */
for o in {mutation, crossover} **do**
  /* Genetic: Mutation or Crossover*/
  $\mathbf{X}'_o$:(B,L,ed) $\leftarrow$ Genetic(SiLU(Conv1D(Linear($\mathbf{X}'_I$))))
  $\mathbf{B}_o$:(B,L,N) $\leftarrow$ Linear^B_o($\mathbf{X}'_I$)
  $\mathbf{C}_o$:(B,L,N) $\leftarrow$ Linear^C_o($\mathbf{X}'_I$)
  /* S6 with Self-Attention */
  /* softplus ensures positive Δ_o */
  Δ_o:(B,L,ed) $\leftarrow$ log(1 + exp(Linear$^\Delta_o$($\mathbf{X}'_o$) + Parameter$^\Delta_o$))
  $\overline{\mathbf{A}}_o$:(B,L,ed,N) $\leftarrow$ $\Delta_o \otimes$ Parameter$^\Delta_o$
  $\overline{\mathbf{B}}_o$:(B,L,ed,N) $\leftarrow$ $\Delta_o \otimes \mathbf{B}_o$
  $\mathbf{H}_{-1}$:(B,L,ed,N) $\leftarrow$ torch.zero()
  $\mathbf{H}_o$:(B,L,ed,N) $\leftarrow$ pscan($\overline{\mathbf{A}}_o \otimes \mathbf{H}_{-1} + \mathbf{B}_o \otimes \mathbf{X}'_o$)
  $\mathbf{H}'_o$:(B,L,d_m) $\leftarrow$ Linear($\mathbf{H}_o$)
  $\mathbf{H}''_o$:(B,L,ed,N) $\leftarrow$ Att($\mathbf{H}'\mathbf{W}_q, \mathbf{H}'\mathbf{W}_k, \mathbf{H}'\mathbf{W}_v$)
  $\mathbf{H}_x$:(B,L,ed,N) $\leftarrow$ Linear($\mathbf{H}''_o$)
  $\mathbf{Y}_o$:(B,L,ed) $\leftarrow \mathbf{H}_x \otimes \mathbf{C}_o$
end for
/* get gated $\mathbf{Y}_o$ */
$\mathbf{X}_m$:(B,L,ed) $\leftarrow \mathbf{Z} \odot \mathbf{Y}_m$
$\mathbf{X}_c$:(B,L,ed) $\leftarrow \mathbf{Z} \odot \mathbf{Y}_c$
$\mathbf{X}_{I+1}$:(B,L,d) $\leftarrow$ Linear$_I$($\mathbf{X}_m + \mathbf{X}_c$)
return $\mathbf{X}_{I+1}$

states to capture contextual information. The ViSTA module facilitates interaction between multiscale visual features and text features. The interacted information is then fed into the text decoder. The details are explained as follows.

1) Visual Encoder: The encoder is composed of a stack of $\mathbb{B}$ identical encoder layers (i.e., GMamba blocks). Each GMamba block stacks a genetic SSM and a position-wise fully connected feedforward network (FFN) in turn. Each sublayer is preceded and followed by a root-mean-square normalization (RMSNorm) [65] and a residual connection, respectively. The input of each GMamba block is the output of the previous GMamba block ($\mathbf{X}_f$ is the input of the first GMamba block). The output $\mathbf{V}^{\text{enc}}_i \in \mathbf{R}^{L\times d}$ of the ith GMamba block is calculated as

$$\mathbf{U}_i = \text{RMSNorm}\big(\text{GSSM}\big(\mathbf{X}^{\text{enc}}_i\big)\big) + \mathbf{X}^{\text{enc}}_i \tag{21}$$

$$\mathbf{V}^{\text{enc}}_i = \text{RMSNorm}(\text{FFN}(\mathbf{U}_i)) + \mathbf{U}_i \tag{22}$$

where RMSNorm is a more efficient variant of layer normalization [66]. Compared to layer normalization, RMSNorm normalizes the inputs of a neuron within a single layer by utilizing the root mean square of those inputs. Then, we can obtain the outputs $\mathbf{V}^{\text{enc}} = \{\mathbf{V}^{\text{enc}}_1, \ldots, \mathbf{V}^{\text{enc}}_{\mathbb{B}}\}$ of the encoder.

2) Vision and Scene Text Aggregation: As shown in Fig. 1, a text sequence is appended with a start token < bos> and an end token < eos>. After positional encoding and word embedding [18], the word sequence $\mathbf{Y} = \{\mathbf{y}_1, \ldots, \mathbf{y}_T\} \in \mathbf{R}^{T\times d}$ is obtained. Drawing inspiration from the application of mesh structures [12], [30], [32] in Transformers to enhance interactions between multilayer visual features and text features, we design a ViSTA module for the Mamba framework to integrate visual and textual information, as outlined in the following:

$$\mathbf{Z}_i = \text{Concat}\big(\text{Mean}\big(\mathbf{V}^{\text{enc}}_i\big), \mathbf{Y}\big) \tag{23}$$

$$\alpha_i = \sigma(\text{Linear}(\mathbf{Z}_i)) \tag{24}$$

$$\mathbf{Y}' = \textstyle\sum_1^{\mathbb{B}} \alpha_i \odot \mathbf{Z}_i \tag{25}$$

where $\mathbf{Z}_i \in \mathbf{R}^{T\times 2d}$, $\alpha_i \in \mathbf{R}^{T\times 2d}$, and $\mathbf{Y}' \in \mathbf{R}^{T\times 2d}$; Mean is the mean operator; Concat means the concatenation operator; and σ is the sigmoid activation function.

3) Text Decoder: Text decoder consists of a stack of $\mathbb{B}$ identical decoder layers (i.e., standard Mamba blocks). Each Mamba block stacks an S6 and an FFN in turn. In line with the encoder, each sublayer employs a pre-RMS normalization followed by a post-residual connection. The input of each Mamba block is the output of the previous Mamba block ($\mathbf{Y}'$ is the input of the first Mamba block). The output $\mathbf{Y}^{\text{dec}}_i \in \mathbf{R}^{T\times d}$ of the ith Mamba block is shown as follows:

$$\Gamma_i = \text{RMSNorm}\big(\text{S6}\big(\mathbf{T}^{\text{dec}}_i\big)\big) + \mathbf{T}^{\text{dec}}_i \tag{26}$$

$$\mathbf{Y}^{\text{dec}}_i = \text{RMSNorm}(\text{FFN}(\Gamma_i)) + \Gamma_i. \tag{27}$$

The output $\mathbf{Y}^{\text{dec}}_{\mathbb{B}}$ of the last Mamba block serves as the decoder's output. Following that, $\mathbf{V}^{\text{dec}}_{\mathbb{B}}$ is passed sequentially through a fully connected layer and a softmax layer to compute the probabilities for a vocabulary of potential words.

F. Training and Objectives

Following previous practices in RSIC [30], [32], we sequentially apply cross-entropy (XE or CE) training and CIDEr-D score optimization with SCST to optimize our model. The details are as follows.

1) Training With Cross-Entropy Loss: In the first training stage, we train our model with a word-level XE loss

$$L_{\text{XE}}(\theta) = -\sum_{t=1}^{T} \log\big(p_\theta\big(y^*_t | y^*_{1:t-1}\big)\big) \tag{28}$$

where $y^*_{1:t-1} = [y^*_1, \ldots, y^*_{t-1}]$ represents a portion of the GT sequence, specifically the sequence of words from the start till the $(t-1)$th time step; T is the maximum time step; and θ denotes the variable parameter of the model.

2) CIDEr-D Score Optimization: Subsequently, we adopt a beam search strategy to select the top-k sentences with the highest probability. Their CIDEr-D scores are utilized as rewards to optimize the model. The gradient expression can be formulated as follows:

$$\nabla_\theta L_{RL}(\theta) = -\frac{1}{k}\sum_{i=1}^{k}\big(\big(r\big(w^i\big) - b\big)\nabla_\theta \log p_\theta\big(w^i\big)\big) \tag{29}$$

where k is the number of samples, w^i denotes the ith sentence sampled in the beam, $r(\cdot)$ represents the reward function, and $b = 1/k\sum_{i=1}^{k} r(w^j)$ is the baseline reward.

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IV. Experiments

A. Datasets and Settings

1) Datasets and Metrics: This article evaluates the proposed MG-Transformer on Sydney-Captions [19], UCM-Captions [20], RSICD [20], and NWPU-Captions [64]. Sydney-Captions comprises 600 RSIs with seven land cover categories, each annotated with five different captions, and features a pixel resolution of 0.5 m. UCM-Captions consists of 2100 RSIs with 21 land cover categories, each annotated with five different captions, and features a pixel resolution of 0.3048 m. RSICD contains 10 921 RSIs with 30 land cover categories, each annotated with less than five different captions, and features various pixel resolutions. NWPU-Captions contains 31 500 RSIs with 45 land cover categories. Each category comprises 700 images, with each RSI annotated with five distinct captions, and the pixel resolution ranges from 30 to 0.2 m. These four datasets are partitioned into training, validation, and testing sets with an 8:1:1 ratio.

For Sydney-Captions, UCM-Captions, and RSICD, most methods [19], [20], [21], [22], [24], [30], [32], [71] rerandomize the training, validation, and testing sets according to the default ratios. Some methods do not explicitly state their dataset partitioning approach [28], [62], [64], [68], [69], [72], while a minority adopt a public partitioning [28], [61], [67]. To ensure a fair comparison, following [30], [32], and [71], we conduct experiments with five different random splits. The highest and lowest results have been discarded, leaving us to present the mean and standard deviation of the remaining outcomes. In addition, we have reproduced the experimental results of RASG [27] and MLAT [31] using the same approach. For NWPU-Captions, we maintain consistency with comparative methods by using the public split.

All methods are evaluated using standard image captioning metrics (BLEU-n [50], METEOR [51], ROUGE_L [52], CIDEr-D [53], SPICE [54], and S_m^* [24]) by employing the officially released codes, where S_m^* is the mean of BLEU-4, METEOR, ROUGE-L, and CIDEr-D.

2) Implementation Details: All RSIs are resized to 224×224. We follow [30] to employ the pretrained CLIP image encoder with ViT-B/32 backbone to extract $L = 49$ patch features with dimensions of 768 (i.e., $d_1 = 768$). Then, the patch features are embedded to a dimension of $d = 512$, which matches the input embedding size of visual encoder and text decoder. We utilize a sequential Mamba framework with $\mathbb{B} = 3$ identical blocks. The expand operator e is set to 2, the hidden state dimension N is 16, and the size of the causal Conv1D kernel is 4. The hidden dimension of each FFN layer is set to 2048. For genetic SSM, the dilation rates of the dilated convolutions in 3 GMamba blocks in the visual encoder are set as 2, 4, and 8 in sequence. The number of chromosomes λ is set to 4, and the number of exchanges c is set to 2. The exchange probability P_c and mutation probability P_m are selected from the range [0, 1]. To ensure a similar number of parameters as comparison methods, the intermediate dimension d_m of the self-attention is set to 32, and the number of attention heads is 1. The text features are obtained using word embedding and positional encoding [18]. The dimension of the word embeddings is 512, the maximum sequence length for positional encoding is 128, and the maximum length for predicting sentences is 20. The batch size B is 50, and the beam size is set to 5. The XE training process first utilizes the AdamW optimizer [55] with $\beta_1 = 0.9$, $\beta_2 = 0.95$, a momentum of 0.9, an initial learning rate of 2e^{-4}, a weight decay of 0.1, and a cosine decay learning rate scheduler. When patience reaches 5, switch to CIDEr-D score optimization, which initializes parameters using the optimal model from the validation set. Unlike XE training, CIDEr-D score optimization adopts the Adam optimizer [56] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and fixed learning rate 5e^{-6}. All experiments are performed on an NVIDIA GeForce RTX 3090 using PyTorch version 2.1.1.

3) Compared Methods: We compare our RSIC-GMamba with 18 RSIC methods based on the CNN-RNN framework, Transformer framework, and Mamba framework. Notably, many large model-based methods also employ the Transformer framework.

1) *CNN-RNN-Based Methods:*
   a) mRNN [19] and mLSTM [19] utilize different CNNs to extract region features and employ different RNNs to decode words.
   b) Soft-attention [20] and Hard-attention [20] establish the connection between the image regions and words based on hard attention [2] and soft attention [2].
   c) FC-Att+LSTM [21] and SM-Att+LSTM [21] inject attribute information or label information into the attention mechanism to promote RSIC.
   d) SAT (LAM) [22] and Adaptive (LAM) [22] design a label-attention mechanism to filter out redundant image information.
   e) GVFGA+LSGA [24] designs a global visual feature-guided attention that filters redundant information through attention gates and proposes linguistic state-guided attention for intermodal interaction.
   f) RASG [27] introduces a recurrent attention mechanism and employs a two-layer LSTM structure to decode captions.
   g) HCNet [61] designs a hierarchical feature aggregation module to combine features of different scales and enhance feature representation and a cross-modal feature interaction module to align visual features with textual features.
   h) MLCANet [62] introduces a multilevel attention module to aggregate multiscale features and a contextual attention mechanism to generate accurate sentences.
2) *Transformer-Based Methods:*
   a) M^2 Transformer [12] models the relationships between objects and designs a meshed cross-attention to establish the interaction between multilayer visual features and text features.
   b) PKG-Transformer [30] models the scene–object relationships and incorporates self-attention to guide the model in generating descriptions that are closely related to the scene.
   c) MLAT [31] integrates multiscale features from multiple convolutional layers and uses LSTMs to consolidate features from various encoding layers.

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TABLE I

COMPARISON RESULTS ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE$_L$, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Methods	B1	B2	B3	B4	M	R	C	S	S_m^*
mRNN [19]	51.30	37.50	20.40	19.30	18.50	-	32.20	-	-
mLSTM [19]	54.60	39.50	22.30	21.20	20.50	-	37.20	-	-
Soft-attention [20]	73.22	66.74	62.23	58.20	39.42	71.27	249.93	-	104.71
Hard-attention [20]	75.91	66.10	58.89	52.58	38.98	71.89	218.19	-	95.41
FC-Att+LSTM [21]	73.83	64.40	57.01	50.85	36.38	66.89	224.15	39.51	94.57
SM-Att+LSTM [21]	74.30	65.35	58.59	51.81	36.41	67.72	234.02	39.76	97.49
SAT(LAM) [22]	74.05	65.50	59.04	53.04	36.89	68.14	235.19	40.38	98.32
Adaptive(LAM) [22]	73.23	63.16	56.29	50.74	36.13	67.75	234.55	42.43	97.29
GVFGA+LSGA [24]	76.81	68.46	61.45	55.04	38.66	70.30	245.22	45.32	102.31
RASG [27]	79.83±2.34	73.71±3.10	68.70±3.89	64.21±4.51	41.29±1.64	72.03±2.57	250.18±11.50	-	106.93±4.99
MLAT [31]	83.23±1.35	77.88±3.99	72.98±4.11	68.29±3.93	43.81±2.11	75.99±2.90	277.24±6.06	-	116.32±2.96
M^2 Transformer [12]	82.25±1.65	76.19±1.12	71.04±1.83	66.30±1.83	44.20±1.49	75.21±1.98	275.44±13.65	43.64±1.74	115.29±2.69
PKG-Transformer [30]	83.17±1.02	77.83±2.31	72.84±1.88	68.24±1.39	45.28±0.86	77.06±1.54	284.76±10.84	44.05±1.05	118.83±2.51
MG-Transformer [32]	**86.68±1.07**	80.87±2.47	75.88±3.54	71.22±4.36	47.70±3.11	79.70±3.86	327.76±14.49	48.09±1.93	131.37±5.19
Mamba-IC	85.00±0.17	79.71±0.49	74.65±1.08	69.85±1.53	46.85±1.46	79.30±0.45	324.74±20.48	47.21±2.84	130.18±5.53
RSIC-GMamba	86.68±0.57	**81.65±0.42**	**77.05±0.89**	**72.48±1.33**	**47.81±0.40**	**80.24±0.23**	**343.57±16.73**	**48.44±0.78**	**136.03±3.72**

d) MG-Transformer [32] combines the CLIP image embedding with an attention gate mechanism to achieve comprehensive image representation and introduces a global grouping attention mechanism to enhance the local modeling capability of the model.

e) PureT [72] is a pure Transformer-based model, which employs Swin Transformer [73] as its backbone to extract region features and incorporates a refining encoder and decoder for generating image captions. It also employs XE training and CIDEr-D score optimization to enhance the quality of generated captions. The experimental results are derived from [64].

f) ClipCap [67] utilizes the pretrained CLIP to extract image features and then generates descriptions through a simple mapping network and a fine-tuned language model GPT-2 [70]. The experimental results are derived from [61].

g) BLIP-2 [68] is a pretrained model for vision–language tasks, which uses the Q-Former structure to establish connections between visual and language models. It undergoes representation learning through three pretraining tasks: image–text matching, image–text contrastive learning, and image–text generation. The results come from [64].

h) RS-CapRet [69] is a vision–language model designed for RSIs, which is trained by combining image captioning tasks with image–text contrastive learning.

i) BITA [64] is a vision–language pretraining method based on a lightweight interactive Fourier Transformer. It conducts image–text contrastive learning on the combined UCM-Captions, RSICD, and NWPU-Captions datasets. Subsequently, it connects the frozen image encoder with a large language model and fine-tunes each dataset to enhance the model performance.

3) *Mamba-Based Method:*

a) Following encoder–decoder paradigm, we design a Mamba baseline for image captioning (Mamba-IC), where both the encoder and text decoder use vanilla Mamba [38]. Its training process is consistent with our RSIC-GMamba.

B. Comparison With Other Methods

The experimental results of all compared methods on the three RSIC datasets are presented as percentages (%) in Tables I–IV. Note that in Tables I–III, we only compare the methods involving random repartitioning for the Sydney-Captions and UCM-Captions datasets. Overall, our method outperforms both the Mamba-IC and the advanced MG-Transformer. The specific analysis results for each RSIC dataset are as follows.

1) Results on Sydney-Captions: As shown in Table I, mRNN and mLSTM struggle to achieve high performance using the vanilla CNN-RNN framework. However, with the integration of attention mechanisms, Soft-attention and Hard-attention significantly enhance the performance of the model, resulting in a more than fivefold increase in the CIDEr-D score. Furthermore, FC-Att+LSTM, SM-Att+LSTM, SAT (LAM), Adaptive (LAM), and GVFGA+LSGA incorporate label or global information into the attention mechanism, which leads to improvements in several metrics. RASG fuses static visual features with multiscale features and incorporates a recurrent attention mechanism, resulting in improvements across all metrics. MLAT, aggregating multilevel visual features through LSTMs, outperforms RASG by 10.82% on CIDEr-D score and 8.78% improvement on S_m^* score. M^2 Transformer encodes prior relationships between objects and performs slightly inferior to MLAT. Subsequently, PKG-Transformer integrates prior scene–object knowledge into self-attention, surpassing MLAT with a 2.71% improvement in CIDEr-D score and a 2.16% improvement in S_m^* score. MG-Transformer, leveraging large visual–language model CLIP image embeddings and global grouping attention mechanism, outperforms PKG-Transformer by 15.10% on CIDEr-D score and 10.55% improvement on S_m^* score. The performance of Mamba-IC, which employs CLIP features, is marginally lower than that of MG-Transformer. RSIC-GMamba integrates the heuristic genetic algorithm and a meshed structure, achieving a 5.80% improvement over Mamba on CIDEr-D score and a 4.49% enhancement on S_m^* score.

2) Results on UCM-Captions: It can be seen from Table II that Transformer-based methods outperform those based

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TABLE II

COMPARISON RESULTS ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE$_L$, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Methods	B1	B2	B3	B4	M	R	C	S	S_m^*
mRNN [19]	60.10	50.70	32.80	20.80	19.30	-	42.80	-	-
mLSTM [19]	63.50	53.20	37.50	21.30	20.30	-	44.50	-	-
Soft-attention [20]	74.54	65.45	58.55	52.50	38.86	72.37	261.24	-	106.24
Hard-attention [20]	81.57	73.12	67.02	61.82	42.63	76.98	299.47	-	120.23
FC-Att+LSTM [21]	81.02	73.30	67.27	61.88	42.80	76.67	337.00	48.67	129.59
SM-Att+LSTM [21]	81.15	74.18	68.14	62.96	43.54	77.93	338.60	48.75	130.76
SAT(LAM) [22]	81.95	77.64	74.85	71.61	48.37	79.08	361.71	50.24	140.19
Adaptive(LAM) [22]	81.7	75.1	69.9	65.4	44.8	78.7	328.0	50.3	129.23
GVFGA+LSGA [24]	83.19	76.57	71.03	65.96	44.36	78.45	332.70	48.53	130.37
RASG [27]	82.05±1.38	77.47±1.30	73.86±1.32	70.85±1.26	47.40±0.36	78.49±1.02	326.01±3.73	-	130.69±1.48
MLAT [31]	90.35±1.04	86.86±1.31	83.68±1.65	80.77±2.11	54.57±1.61	87.67±1.46	383.15±17.52	-	151.54±5.60
M^2 Transformer [12]	88.90±1.32	85.98±1.49	82.74±1.51	80.89±1.36	51.13±2.21	84.45±2.61	418.41±17.27	53.43±2.43	155.97±4.36
PKG-Transformer [30]	90.48±1.15	87.04±1.09	84.10±1.33	81.39±1.67	**54.66±2.05**	86.57±2.00	427.49±12.37	57.01±2.40	162.53±3.52
MG-Transformer [32]	91.28±1.23	88.11±2.29	85.37±2.17	82.95±2.08	56.01±1.66	**88.77±2.06**	448.39±4.08	**58.39±1.32**	169.03±2.05
Mamba-IC	89.89±0.59	86.42±0.71	83.50±0.73	80.80±0.61	53.60±0.71	87.23±0.58	436.66±3.67	56.39±0.53	164.72±0.98
RSIC-GMamba	**91.40±1.12**	**88.42±1.30**	**85.72±1.43**	**83.18±1.62**	54.65±0.56	88.22±0.47	**451.65±6.68**	57.17±0.37	**169.42±1.15**

TABLE III

COMPARISON RESULTS ON RSICD. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE$_L$, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Methods	B1	B2	B3	B4	M	R	C	S	S_m^*
mRNN [19]	45.58	28.25	18.09	12.13	15.69	31.26	19.15	-	19.56
mLSTM [19]	50.57	32.42	23.19	17.46	17.84	35.02	31.61	-	25.48
Soft-attention [20]	67.53	53.08	43.33	36.17	32.55	61.09	196.43	-	
Hard-attention [20]	66.69	51.82	41.64	34.07	32.01	60.84	179.25	-	76.54
FC-Att+LSTM [21]	66.71	55.11	46.91	40.59	32.25	57.81	257.63	46.73	97.07
SM-Att+LSTM [21]	66.99	55.23	47.03	40.68	32.55	58.02	257.38	46.87	97.16
SAT(LAM) [22]	67.53	55.37	46.86	40.26	32.54	58.23	258.50	46.36	97.38
Adaptive(LAM) [22]	66.64	54.86	46.76	40.70	32.30	58.43	260.55	46.73	98.00
GVFGA+LSGA [24]	67.79	56.00	47.81	41.65	32.85	59.29	260.12	46.83	98.48
RASG [27]	69.57±1.39	57.24±0.42	47.10±0.52	39.75±0.60	34.00±0.24	**64.48±0.42**	247.09±3.44	-	96.33±1.10
MLAT [31]	69.46±1.06	59.20±1.25	51.02±1.32	44.69±1.30	33.95±0.50	61.81±1.22	269.75±3.70	-	102.55±1.39
M^2 Transformer [12]	68.44±1.20	56.57±1.25	48.10±1.25	41.56±1.26	32.69±1.01	59.12±0.93	258.58±3.68	45.43±0.92	97.99±1.67
PKG-Transformer [30]	69.67±1.74	58.30±1.39	50.45±1.13	44.31±1.24	33.32±1.31	60.78±1.30	274.01±2.88	46.91±0.79	103.11±1.31
MG-Transformer [32]	70.27±1.54	**59.80±0.24**	**52.08±0.46**	**46.01±0.65**	**34.17±0.52**	62.27±0.93	**285.10±5.01**	**48.07±0.98**	**107.89±1.78**
Mamba-IC	69.21±1.02	58.54±0.76	50.80±0.77	44.71±0.92	33.33±0.24	60.92±0.52	270.80±2.54	46.68±0.27	102.44±1.03
RSIC-GMamba	69.38±0.76	59.04±0.11	51.42±0.41	45.38±0.75	33.63±0.40	61.73±0.88	278.13±6.16	47.43±1.05	104.72±1.96

CNN-RNN framework. Specifically, the optimal Transformer-based method, MG-Transformer, surpasses the CNN-RNN-based method SAT (LAM) by 20.96% and 20.57% on CIDEr-D score and S_m^* score, respectively. Mamba-IC significantly outperforms above CNN-RNN methods. However, it lags behind MG-Transformer, exhibiting a 2.69% lower CIDEr-D score and a 2.62% lower S_m^* score. RSIC-GMamba exceeds Mamba-IC by 3.43% and 2.85% on CIDEr-D score and S_m^* score.

3) Results on RSICD: As shown in Table III, Transformer-based methods outperform CNN-RNN-based methods. The performance achieved by Mamba-IC exceeds MLAT and M^2 Transformer but is lower than MG-Transformer and PKG-Transformer. Our RSIC-GMamba surpasses Mamba-IC by 2.71% and 2.23% on CIDEr-D score and S_m^* score, respectively, reducing the gap with the best MG-Transformer. This could be attributed to the fact that RSICD exhibits a wider range of pixel resolutions compared to Sydney-Captions and UCM-Captions. In addition, MG-Transformer integrates region features and CLIP image embedding, resulting in more robust visual feature representations.

4) Results on NWPU-Captions: As shown in Table IV, Soft-attention, Hard-attention, FC-Att+LSTM, SM-Att+LSTM, and MLCA-Net achieve acceptable results by integrating attention mechanisms, attribute information, and multiscale information within the CNN-RNN framework. Subsequently, HCNet adopts a dual LSTM structure, significantly improving the CIDEr-D score and S_m^* score through hierarchical feature aggregation and cross-modal feature alignment. Transformer-based methods also demonstrate outstanding performance. Specifically, compared to CNN-RNN-based methods, MLAT also utilizes a CNN pretrained on ImageNet to extract visual features, achieving performance that is second only to HCNet. RS-CapRet, Clipcap, BLIP-2, BITA, PureT, and MG-Transformer further fine-tune lightweight Transformers based on prior knowledge from large vision models or vision–language models to enhance the performance. MG-Transformer achieved the best performance, surpassing HCNet by 3.45% and 0.48% in the CIDEr-D score and S_m^* score, respectively. Compared to the aforementioned methods, the novel Mamba-IC exhibits performance in BLEU-2, CIDEr-D, and S_m^* that is only

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TABLE IV

COMPARISON RESULTS ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE$_L$, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. SYMBOL "-" INDICATES THAT THE RESULT IS NOT REPORTED BY THIS ARTICLE

Methods	B1	B2	B3	B4	M	R	C	S	S_m^*
Soft-attention [20]	73.10	60.90	52.50	46.20	33.90	59.90	113.60	-	63.40
Hard-attention [20]	73.30	61.00	52.70	46.40	34.00	60.00	110.30	-	62.67
FC-Att+LSTM [21]	73.60	61.50	53.20	46.90	33.80	60.00	123.10	-	65.95
SM-Att+LSTM [21]	73.90	61.70	53.20	46.80	33.00	59.30	123.60	-	65.67
MLCA-Net [62]	74.5	62.4	54.1	47.8	33.7	60.1	116.4	28.5	64.5
HCNet [61]	89.54	82.51	76.58	71.68	47.42	80.53	209.27	-	102.22
RS-CapRet [69]	87.10	78.70	71.70	65.60	43.60	77.60	192.90	31.10	94.93
Clipcap [67]	83.94	74.21	66.19	59.54	41.42	73.85	172.65	-	86.85
BLIP-2 [68]	87.78	80.24	73.51	67.62	45.01	78.44	193.86	33.28	96.23
BITA [64]	88.54	80.70	73.76	67.60	45.27	78.53	197.04	33.65	97.11
PureT [72]	88.82	80.31	73.30	67.50	42.32	75.84	195.12	27.43	95.20
MLAT [31]	85.27	76.74	70.07	64.79	43.27	74.96	185.56	-	92.14
MG-Transformer [32]	91.14	84.00	77.78	72.46	43.46	78.45	216.48	27.26	102.71
Mamba-IC	91.14	81.71	73.03	65.65	38.61	72.62	203.35	24.63	95.06
RSIC-GMamba	**92.43**	**85.71**	**79.80**	**74.71**	**45.37**	**80.32**	**223.49**	**29.03**	**105.97**

TABLE V

ABLATION STUDIES ON SYDNEY-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE$_L$, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. DC, CB, MB, AND SA REPRESENT DILATED CONVOLUTION, CROSSOVER BRANCH, MUTATION BRANCH, AND SELF-ATTENTION, RESPECTIVELY

ViSTA	DC	CB	MB	SA	B1	B2	B3	B4	M	R	C	S	S_m^*
✗	✗	✗	✗	✗	85.00±0.17	79.71±0.49	74.65±1.08	69.85±1.53	46.85±1.46	79.30±0.45	324.74±20.48	47.21±2.84	130.18±5.53
✓	✗	✗	✗	✗	86.58±0.44	80.94±1.32	75.51±2.31	70.37±3.03	46.18±0.80	79.44±0.63	332.72±11.27	47.36±0.89	132.18±2.97
✓	✓	✗	✗	✗	86.63±0.52	81.22±1.15	76.08±2.16	71.10±2.83	46.52±1.67	79.68±1.22	333.73±11.73	47.19±0.45	132.76±1.94
✓	✓	✓	✗	✗	86.43±0.87	81.14±1.40	76.03±2.16	70.97±2.67	47.61±1.47	79.93±1.17	336.56±24.97	47.90±1.86	133.77±6.73
✓	✓	✗	✓	✗	86.56±1.16	81.57±1.52	76.72±2.06	71.99±2.47	46.99±1.12	79.98±0.75	336.02±16.02	47.79±1.59	133.74±4.31
✓	✓	✓	✓	✗	86.43±1.29	81.12±1.49	76.01±2.19	71.30±2.65	47.10±1.12	79.64±0.68	339.01±18.20	48.03±2.19	134.26±4.88
✓	✓	✓	✓	✓	**86.68±0.57**	**81.65±0.42**	**77.05±0.89**	**72.48±1.33**	**47.81±0.40**	**80.24±0.23**	**343.57±16.73**	**48.44±0.78**	**136.03±3.72**

TABLE VI

ABLATION STUDIES ON UCM-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE$_L$, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. DC, CB, MB, AND SA REPRESENT DILATED CONVOLUTION, CROSSOVER BRANCH, MUTATION BRANCH, AND SELF-ATTENTION, RESPECTIVELY

ViSTA	DC	CB	MB	SA	B1	B2	B3	B4	M	R	C	S	S_m^*
✗	✗	✗	✗	✗	89.89±0.59	86.42±0.71	83.50±0.73	80.80±0.61	53.60±0.71	87.23±0.58	436.66±3.67	56.39±0.53	164.72±0.98
✓	✗	✗	✗	✗	90.29±1.37	86.97±1.37	84.08±1.40	81.25±1.48	53.82±0.94	87.41±0.37	440.73±5.44	56.88±0.62	165.98±1.63
✓	✓	✗	✗	✗	90.61±0.97	87.18±1.11	84.11±1.05	81.43±1.05	54.34±0.74	87.42±0.64	442.59±4.18	56.91±0.64	166.21±0.69
✓	✓	✓	✗	✗	91.04±0.90	87.80±0.96	84.77±0.85	82.42±0.74	54.43±0.35	87.85±0.43	445.39±5.10	57.01±0.50	167.52±1.47
✓	✓	✗	✓	✗	91.04±0.22	87.70±0.41	84.82±0.48	82.17±0.52	54.70±0.84	87.88±0.63	446.52±5.38	56.89±1.43	167.82±1.61
✓	✓	✓	✓	✗	91.32±0.75	87.91±0.80	84.96±0.89	82.85±1.07	54.65±0.62	87.97±0.26	447.92±5.80	57.17±0.54	167.79±1.36
✓	✓	✓	✓	✓	**91.40±1.12**	**88.42±1.30**	**85.72±1.43**	**83.18±1.62**	**54.82±0.56**	**88.22±0.47**	**451.65±6.68**	**57.53±0.37**	**169.42±1.15**

inferior to HCNet and MG-Transformer. In comparison to the best-performing MG-Transformer, our RSIC-GMamba achieves the best performance across all metrics, particularly improving the CIDEr-D score and S_m^* score by 3.24% and 3.17%, respectively.

The analysis above demonstrates that our RSIC-GMamba not only surpasses the performance of the baseline Mamba-IC model but also outperforms numerous advanced Transformer models for RSIC.

C. Ablation Study

In this section, we conduct extensive and detailed ablation experiments on Sydney-Captions, UCM-Captions, RSICD, and NWPU-Captions to evaluate the effectiveness of the ViSTA module and the genetic SSM, which is composed of dilated convolution (DC), mutation branch (MB), cross branch (CB), and self-attention (SA). To further validate the superiority of the genetic SSM, we perform replacement experiments using various SSMs from SOTA visual Mamba variants to replace the genetic SSM of RSIC-GMamba. In addition, we investigate the effects of P_m and P_c for genetic SSM. Finally, we compared the performance of XE training and RL training.

1) Effectiveness of ViSTA and Genetic SSM: Each component is added sequentially to Mamba-IC. All experimental results on the three RSIC datasets are presented in Tables V–VIII. These results reveal a consistent improvement in model performance with the progressive addition of components. For instance, Table VIII illustrates that incorporating ViSTA results in a 5.77% increase in

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TABLE VII

ABLATION STUDIES ON RSICD. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE_L, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. DC, CB, MB, AND SA REPRESENT DILATED CONVOLUTION, CROSSOVER BRANCH, MUTATION BRANCH, AND SELF-ATTENTION, RESPECTIVELY

ViSTA	DC	CB	MB	SA	B1	B2	B3	B4	M	R	C	S	S_m^*
✗	✗	✗	✗	✗	69.21±1.02	58.54±0.76	50.80±0.77	44.71±0.92	33.33±0.24	60.92±0.52	270.80±2.54	46.68±0.27	102.44±1.03
✓	✗	✗	✗	✗	69.82±0.12	58.94±0.24	51.00±0.23	44.81±0.30	33.75±0.47	61.20±0.41	272.60±4.74	47.29±0.69	103.09±1.26
✓	✓	✗	✗	✗	**70.11±0.6**	59.13±0.56	51.06±0.59	44.76±0.67	33.93±0.40	61.62±0.59	273.46±2.53	47.49±0.63	103.44±0.96
✓	✓	✓	✗	✗	69.99±0.58	**59.23±0.38**	51.27±0.46	45.03±0.69	**33.95±0.30**	61.57±0.68	275.39±4.74	**47.51±1.16**	103.99±1.44
✓	✓	✗	✓	✗	69.87±0.76	59.05±0.59	51.17±0.47	44.97±0.44	33.74±0.24	61.76±0.34	277.14±0.86	47.53±0.35	104.40±0.21
✓	✓	✓	✓	✗	69.53±0.70	59.12±0.16	**51.42±0.41**	45.32±0.69	33.64±0.40	**61.75±0.87**	277.57±6.62	47.36±1.12	104.57±2.05
✓	✓	✓	✓	✓	69.38±0.76	59.04±0.11	**51.42±0.41**	**45.38±0.75**	33.63±0.40	61.73±0.88	**278.13±6.16**	47.43±1.05	**104.72±1.96**

TABLE VIII

ABLATION STUDIES ON NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE_L, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY. DC, CB, MB, AND SA REPRESENT DILATED CONVOLUTION, CROSSOVER BRANCH, MUTATION BRANCH, AND SELF-ATTENTION, RESPECTIVELY

ViSTA	DC	CB	MB	SA	B1	B2	B3	B4	M	R	C	S	S_m^*
✗	✗	✗	✗	✗	91.14	81.71	73.03	65.65	38.61	72.62	203.35	24.63	95.06
✓	✗	✗	✗	✗	91.23	83.56	76.88	70.87	42.31	77.04	215.08	27.89	101.33
✓	✓	✗	✗	✗	91.46	83.98	77.44	71.83	43.76	78.35	216.54	28.76	102.62
✓	✓	✓	✗	✗	91.83	84.70	78.37	72.76	43.84	78.76	219.24	28.94	103.65
✓	✓	✗	✓	✗	91.75	84.56	78.18	72.86	44.52	79.59	219.21	27.56	104.04
✓	✓	✓	✓	✗	91.89	84.89	78.61	73.32	45.25	80.01	221.74	28.41	105.08
✓	✓	✓	✓	✓	**92.43**	**85.71**	**79.80**	**74.71**	**45.37**	**80.32**	**223.49**	**29.03**	**105.97**

CIDEr-D score and a 6.60% increase in S_m^* score. This enhancement can be attributed to the effective interaction between multilayer visual features and text features, which allows the model to capture more detailed semantic correspondences. Furthermore, the integration DCs with varying dilation rates in SSM layers enhances the model's ability to capture multiscale objects. This addition leads to a 0.68% boost in CIDEr-D score and a 1.27% boost in S_m^* score. Subsequently, the individual addition of CB improved the capture of richer visual context dependencies, contributing to a 1.25% increase in CIDEr-D score and a 1.00% increase in S_m^* score. In contrast, the individual addition of MB enhanced the model's generalization ability by introducing randomness, leading to a 1.23% increase in the CIDEr-D score and a 1.38% increase in S_m^* score. The combined use of CB and MB improves the CIDEr-D score and the S_m^* score by 2.40% each. In the end, the incorporation of SA, which models the relationships between hidden states in the SSM, leads to the most significant gains, with CIDEr-D score increasing by 0.79% and S_m^* score by 0.85%. Compared to the addition of ViSTA, further integration of genetic SSM results in additional increases of 3.91% in CIDEr-D score and 4.58% in S_m^* score.

Overall, our method exhibits a marked improvement over Mamba-IC, substantially enhancing the applicability of the Mamba framework for RSIC. Specifically, on the Sydney-Captions dataset, the CIDEr-D score and S_m^* score show relative increases of 5.80% and 4.49%, respectively. Similarly, on the UCM-Captions dataset, the improvements are 3.43% for CIDEr-D score and 2.85% for S_m^* score. Notably, on the RSICD dataset, our method achieves enhancements of 2.71% in CIDEr-D score and 2.23% in S_m^* score. Finally, on the NWPU-Captions dataset, our method achieves enhancements of 9.90% in CIDEr-D score and 11.48% in S_m^* score.

2) Replacement Experiment of Genetic SSM With SSMs From SOTA Visual Mamba Variants: We select Vim [39] and MambaMIL [57] as baselines, both featuring two SSM branches. The genetic SSM is replaced with the bidirectional SSM of Vim, defined as RSIC-GMamba-B, and with the sequence reordering SSM from MambaMIL, defined as RSIC-GMamba-S. The experimental results are shown in Table IX. From the table, it can be observed that RSIC-GMamba-B slightly outperforms RSIC-GMamba-S on Sydney-Captions and UCM-Captions, while its performance on the RSICD dataset is marginally lower than RSIC-GMamba-S. Our RSIC-GMamba significantly outperforms both RSIC-GMamba-B and RSIC-GMamba-S across all datasets. For instance, on the UCM dataset, RSIC-GMamba achieves performance improvements of 0.67% and 0.40% over RSIC-GMamba-B for CIDEr-D score and S_m^* score, respectively. This is primarily attributed to the genetic SSM's larger 2-D scanning space and its enhanced generalization capability.

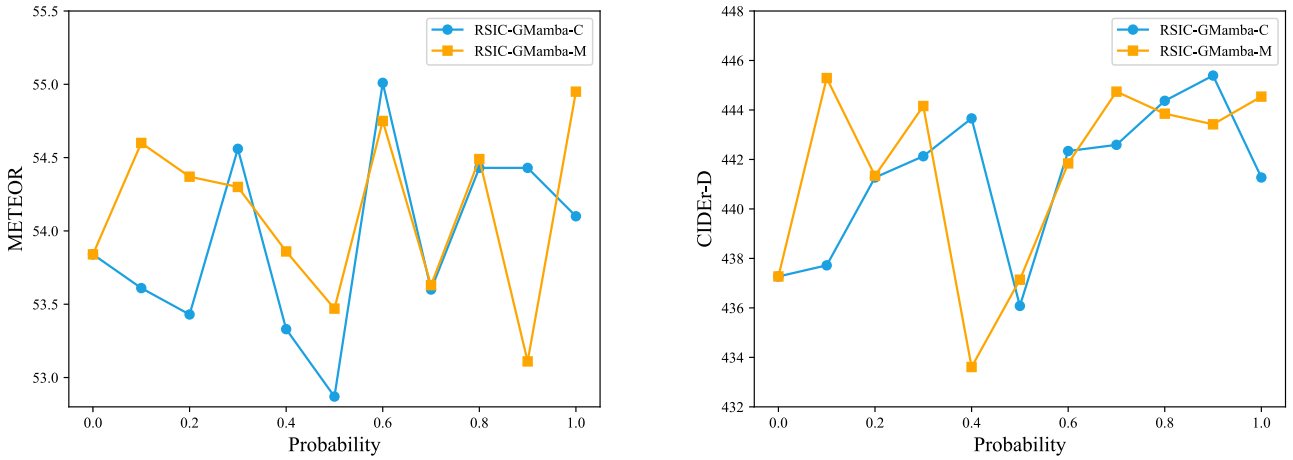
3) Effects of P_m and P_c: To further analyze the effects of CB and MB for genetic SSM, we define RSIC-GMamba-C as RSIC-GMamba without mutation operation and RSIC-GMamba-M as RSIC-GMamba without the crossover operation. RSIC-GMamba-C and RSIC-GMamba-M separately set crossover probability P_m and mutation probability P_c to 0, resulting in CB and MB maintaining original patch sequence order. Subsequently, we conduct experiments on the UCM-Captions with varying values of P_c and P_m. The variation curves of METEOR and CIDEr-D with different values of P_c and P_m are presented in Fig. 3. Intuitively, when the values of P_c or P_m are set to 0.4 or 0.5, the METEOR and CIDEr-D scores reach their lowest points. Conversely, when the values of P_c or P_m exceed 0.5, the METEOR and CIDEr-D

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TABLE IX

EXPERIMENTAL RESULTS OF RSIC-GMAMBA, RSIC-GMAMBA-B, AND RSIC-GMAMBA-S ON SYDNEY-CAPTIONS, UCM-CAPTIONS, RSICD, AND NWPU-CAPTIONS. ALL RESULTS ARE REPORTED AS PERCENTAGE (%). BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE_L, CIDER-D, AND SPICE ARE DENOTED AS B1, B2, B3, B4, M, R, C, AND S, RESPECTIVELY

Datasets	Method	B1	B2	B3	B4	M	R	C	S	S_m^*
Sydney-Captions	RSIC-GMamba-B	85.84±1.70	80.54±2.22	75.60±3.05	70.89±3.96	47.51±1.87	80.18±1.33	326.01±24.55	47.87±1.50	131.15±5.95
	RSIC-GMamba-S	85.50±1.55	80.14±2.52	75.20±3.40	70.55±4.10	46.52±2.10	79.05±1.84	326.09±26.11	47.14±1.87	130.55±8.36
	RSIC-GMamba	**86.68±0.5**	**81.65±0.42**	**77.05±0.89**	**72.48±1.33**	**47.81±0.40**	**80.24±0.23**	**343.57±16.73**	**48.44±0.78**	**136.03±3.72**
UCM-Captions	RSIC-GMamba-B	90.98±0.23	87.63±0.09	84.61±0.07	81.84±0.15	**55.75±0.86**	**88.77±0.58**	448.63±7.56	**58.29±1.76**	168.75±2.24
	RSIC-GMamba-S	90.88±0.36	87.82±0.40	85.03±0.42	82.39±0.38	55.08±1.09	88.07±0.39	445.76±9.33	57.16±1.17	167.83±2.61
	RSIC-GMamba	**91.40±1.12**	**88.42±1.30**	**85.72±1.43**	**83.18±1.62**	54.82±0.56	88.22±0.47	**451.65±6.68**	57.53±0.37	**169.42±1.15**
RSICD	RSIC-GMamba-B	**70.15±1.05**	**59.21±0.98**	51.23±0.95	45.01±1.01	33.54±0.58	61.16±0.52	274.02±1.76	47.18±0.35	103.43±0.91
	RSIC-GMamba-S	70.40±0.83	59.31±1.22	51.30±1.46	45.06±1.68	**33.77±0.58**	61.22±1.11	274.37±4.45	47.16±0.39	103.61±1.91
	RSIC-GMamba	69.38±0.76	59.04±0.11	**51.42±0.41**	**45.38±0.75**	33.63±0.40	**61.73±0.88**	**278.13±6.16**	**47.43±1.05**	**104.72±1.96**
NWPU-Captions	RSIC-GMamba-B	91.82	84.24	77.55	71.73	43.65	78.31	217.39	28.29	102.77
	RSIC-GMamba-S	92.01	84.99	78.80	73.61	45.12	79.91	219.37	28.41	104.50
	RSIC-GMamba	**92.43**	**85.71**	**79.80**	**74.71**	**45.37**	**80.32**	**223.49**	**29.03**	**105.97**



(a) (b)

Fig. 3. Variation on (a) METEOR and (b) CIDEr-D with different probability values. RSIC-GMamba-C denotes RSIC-GMamba without the mutation operation (i.e., $P_m = 0$). RSIC-GMamba-M denotes RSIC-GMamba without the crossover operation (i.e., $P_c = 0$).

scores increase significantly. In most instances, the results for P_c or P_m surpass those obtained with $P_c = 0$ or $P_m = 0$. This further suggests that both MB and CB play a crucial role in enhancing the model's performance.

4) Comparison of the Performance Between XE Training and RL Training: Taking NUPU-Captions as an example, we present the changes in various metrics for XE training and RL training in Fig. 4. From this figure, it is evident that in comparison to the metrics achieved through XE training, all metrics have been enhanced following RL training, with the CIDEr-D score exhibiting the most substantial improvement. This is consistent with findings in NIC tasks [10], [11], [12].

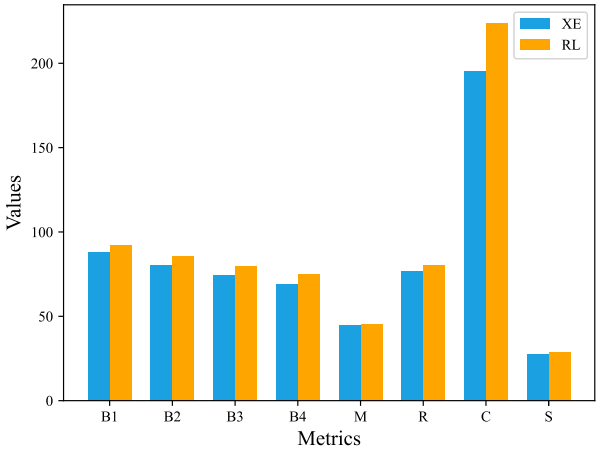


Fig. 4. Comparative analysis of XE and RL training metrics on NWPU–Captions.

D. Qualitative Analysis and Visual Visualization

1) Qualitative Analysis: To intuitively demonstrate the performance of the proposed RSIC-GMamba, qualitative results of Mamba-IC, MG-Transformer, and the proposed Mamba-IC, coupled with human-annotated GT captions, are presented in Fig. 5. Results show that our RSIC-GMamba generates more accurate captions compared to the Mamba-IC and MG-Transformer. For example, in the first subfigure of Fig. 5, both Mamba-IC and MG-Transformer generate the same sentence describing the relationship between numerous mobile homes and mobile home parks, but they omit the description of the road. In addition, in the second subfigure of Fig. 5, Mamba-IC describes a straight runway and some marking

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Fig. 5. Examples of captions generated by the Mamba-IC, MG-Transformer, and the proposed RSIC-GMamba, as well as the corresponding GT captions. Some detailed and accurate words are marked in blue.

lines, while MG-Transformer further specifies that the color of the marking lines is white. Our RSIC-GMamba captures additional details, including a number, which is closer to the GT. This indicates that our RSIC-GMamba is capable of producing more accurate descriptions than Mamba-IC and MG-Transformer.

2) Visual Visualization: To demonstrate the effectiveness of RSIC-GMamba in capturing 2-D visual context, we present the heatmaps of the visual encoder layers for Mamba-IC, RSIC-GMamba-B, and RSIC-GMamba in Fig. 6. Taking the first subfigure in Fig. 6 as an example, the first layer of Mamba-IC focuses solely on the swimming pool, while the second and third layers pay attention to parts of the house and plants. In contrast, the heatmap of RSIC-GMamba-B shows that the red regions across all layers are concentrated around the swimming pool, lacking attention to other targets. Beyond the swimming pool, our RSIC-GMamba model comprehensively focuses on all houses and plants. This observation suggests that our RSIC-GMamba may be better at modeling contextual relationships than Mamba-IC and RSIC-GMamba-B.

E. Complexity Analysis

This section analyzes the computational complexity of the proposed RSIC-GMamba and compares its number of model parameters and number of floating-point operations (FLOPs), as well as running time (i.e., training time per epoch and inference time) with other methods (i.e., PureT, BLIP-2, BITA, MG-Transformer, and Mamba-IC). The encoder of RSIC-GMamba incorporates a lightweight self-attention layer into the SSM layer, preserving a computational complexity of $O(N)$. The text decoder of RSIC-GMamba adopts the vanilla Mamba framework, maintaining a linear inference speed with a complexity of $O(N)$. Therefore, the overall computational complexity of RSIC-GMamba is $O(N)$.

As shown in Table X, MG-Transformer, Mamba-IC, and RSIC-GMamba have comparable model parameters, which is significantly lower than that of PureT, BLIP-2, and BITA. To mitigate the impact of the experimental environment on the running time, MG-Transformer, Mamba-IC, and RSIC-GMamba adopt the experimental settings proposed in this article. Compared to MG-Transformer, Mamba-IC and RSIC-GMamba demonstrate a significant reduction in training time per epoch and inference time. This is primarily due to the strong linear training and inference capabilities of SSM.

In addition, we conduct a detailed analysis of the impact of each component. Compared to Mamba-IC, Mamba-IC+ViSTA shows slight improvements across all metrics. Due to the incorporation of multiple dilated convolutions, Mamba-IC+ViSTA+DC exhibits a significant increase in both model

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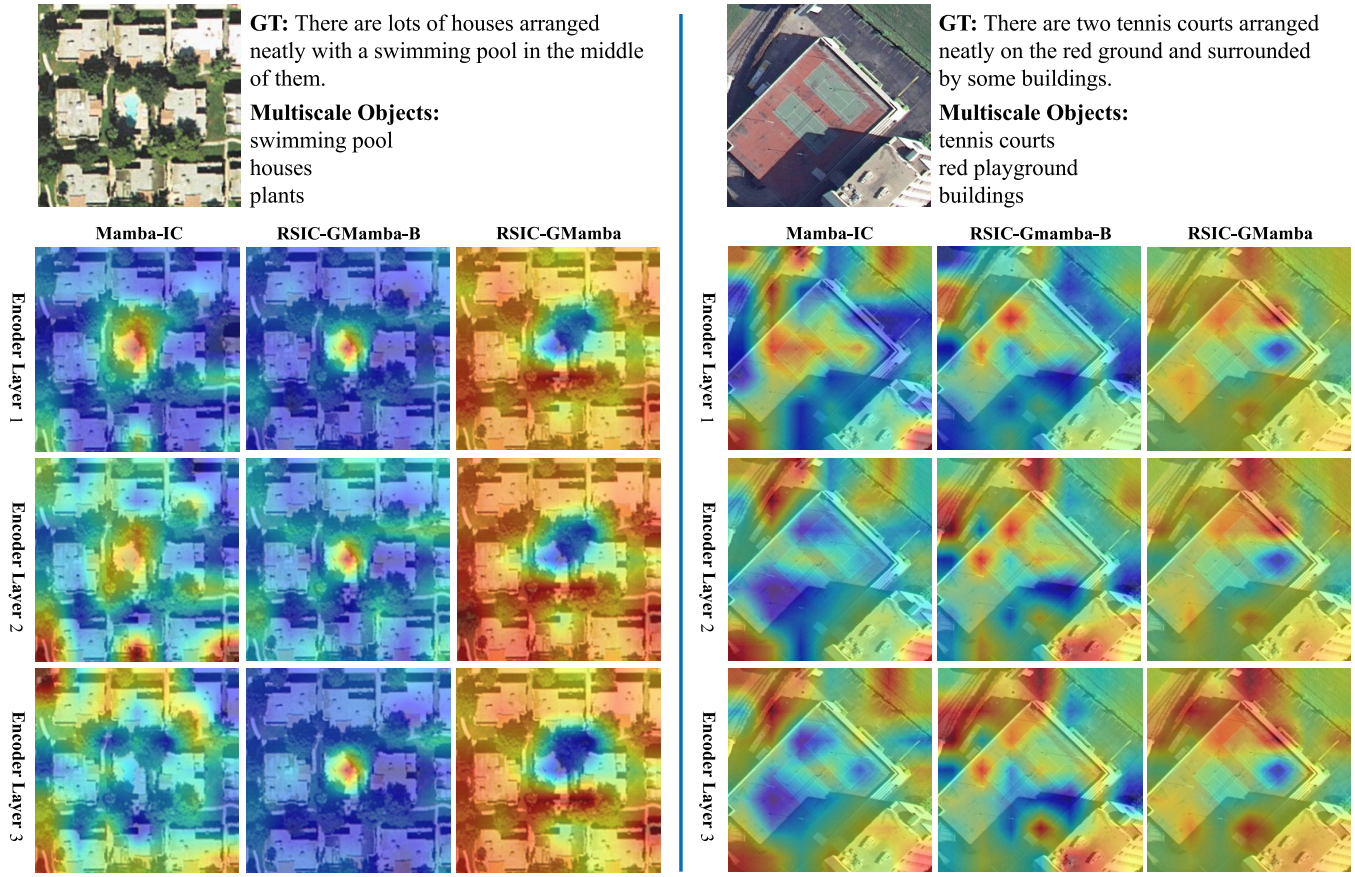


Fig. 6. Comparison of heatmaps generated by all encoding layers of Mamba-IC, RSIC-GMamba-B, and RSIC-GMamba. Red indicates a higher level of attention, while blue represents a lower level of attention.

TABLE X

COMPARISON OF THE PROPOSED RSIC-GMAMBA WITH OTHER METHODS IN TERMS OF THE NUMBER OF MODEL PARAMETERS, FLOPS, TRAINING TIME PER EPOCH, AND INFERENCE TIME ON NWPU-CAPTIONS

Method	Parameters (M)	FLOPs (G)	Training time per epoch (min)	Inference Time (sec)
PureT [72]	220	-	-	-
BLIP-2 [68]	188	-	-	-
BITA [64]	171	-	-	-
MG-Transformer [32]	38.56	1.60	83	20
Mamba-IC	23.78	0.80	5.67	6
Mamba-IC+ViSTA	25.35	0.83	7.07	7
Mamba-IC+ViSTA+DC	32.43	1.17	7.23	7
Mamba-IC+ViSTA+DC+CB	32.80	1.17	9.38	8
Mamba-IC+ViSTA+DC+MB	32.80	1.17	7.52	7
Mamba-IC+ViSTA+DC+CB+MB	32.80	1.18	11.90	9
RSIC-GMamba	39.21	1.50	12.12	10

parameters and FLOPs. The model parameters and FLOPs of Mamba-IC+ViSTA+DC+CB, Mamba-IC+ViSTA+DC+MB, and Mamba-IC+ViSTA+DC+CB+MB exhibit minimal variation. However, the inclusion of CB leads to an increase in both training and inference times. With the addition of self-attention, the performance of RSIC-GMamba is further improved.

V. CONCLUSION

This article presents a novel Mamba-based framework named RSIC-GMamba for RSIC. We develop a genetic SSM that integrates genetic operations and self-attention mechanisms into SSM to fully capture the contextual dependencies of 2-D RSIs. In addition, we implement a ViSTA module for Mamba framework to facilitate interaction between multilayer visual features and text features. Experimental results demonstrate that our method greatly enhances the performance of vanilla Mamba and outperforms many advanced Transformer-based methods. The proposed method has two main limitations.

1) The genetic operations are relatively simplistic and lack domain-specific adaptability for vision–language tasks.
2) There is insufficient consideration of accurate alignment between remote sensing visual and textual semantics. Future research should focus on designing task-aware

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fitness functions that integrate semantic relevance metrics and thoroughly explore the development and optimization of remote sensing vision–language models.

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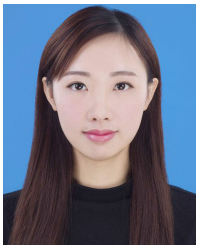
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